Object-Centric Vision-Language Programs for Few-Shot Spatial Concept Induction

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***Abstract*—Vision-language models (VLMs) describe images fluently, yet they often fail to induce a consistent rule from a few labeled examples. Vision-Language Programs (VLP) address this by letting a VLM list the objects it sees and searching for an executable program over those symbols, but every object is text only, so rules about where objects are cannot be written. We give VLP object positions: the same VLM returns every object with a bounding box, and the program grammar gains counting, region and relation predicates, added with a small prior so that the search still tries VLP's own rules first. With one open VLM (Qwen2.5-VL-7B) on CLEVR-Hans7, balanced accuracy on unconfounded test images rises from 69.5% to 80.2% over 21 tasks (exact task-level permutation test, $p = 0.013$), and from 65.6% to 90.8% on the three classes defined by spatial rules ($p = 0.008$), while accuracy on the other classes is unchanged. Feeding images at twice their size makes the VLM's boxes reliable enough for this: the front-to-back order of two objects matches the ground truth 98% of the time. Explicit object geometry is therefore a cheap, training-free way to extend program-based visual reasoning to spatial concepts.**

***Index Terms*—vision-language models, program synthesis, neuro-symbolic reasoning, spatial reasoning, few-shot concept learning**

I. Introduction

Few-shot visual concept induction asks a system to look at a handful of images that follow a hidden rule and a handful that do not, state the rule, and apply it to new images, in the tradition of Bongard problems [1]–[4]. Modern VLMs perceive almost any image, but they reason poorly over what they see: on the original Bongard problems the best VLM solved 43 of 100 puzzles against 68 for the best human [5], and on controlled tests of basic relations such as *left of* or *on* a strong VLM reached 56% against 99% for humans [6], [7].

Vision-Language Programs (VLP) [8] separate the two jobs. The VLM only describes each image as a list of objects with their properties; a program search then finds an executable rule over those symbols that agrees with the labeled examples. On CLEVR-Hans3 [9] this raised the accuracy of Qwen2.5-VL-7B from 46.1% to 86.1%. Every object in VLP, however, is a name plus some property words, and the grammar has no predicate for an object's position or for a relation between two objects. Rules that depend on arrangement, such as *three spheres in the left half* or *a small cyan object in front of two red objects*, cannot be expressed at all. The authors list the missing explicit object representations as a limitation [8], and their synthetic benchmark cannot reveal it, because none of the three CLEVR-Hans3 rules is spatial.

This paper closes that gap with three changes, each tested one at a time:

- **Object-centric symbols.** The same VLM returns every object with a bounding box. The box centre is attached to the object as an attribute outside its word list, so VLP's vocabulary and functions see exactly what they saw before.
- **A spatial grammar.** Counting, region and relation predicates are added to VLP's grammar, with a prior weight that keeps VLP's own rules cheap for the search.
- **An evaluation that can see spatial rules.** We test on CLEVR-Hans7, whose classes 3, 5 and 6 are spatial, and score every task twice: on the confounded validation queries of VLP's protocol and on the unconfounded test split, where only the true rule pays.

Fig. 1 shows the result on one test image: the VLM's own boxes become positioned symbols, and the search finds the class rule `exists_related(cyan, in_front_of, red)`, which VLP's language cannot express.

II. Related Work

Programs for visual reasoning. Neuro-symbolic systems such as NS-VQA [10] and NS-CL [11] parse a scene into objects and execute programs over them, but their perception modules are trained for one synthetic domain such as CLEVR [12]. VisProg [13] and ViperGPT [14] let a language model write a program for an explicit question, which does not apply when the rule itself is unknown, and VPD [15] distils such programs back into one VLM, losing the inspectable program. VLP [8] induces programs from labeled images with an off-the-shelf VLM as perception; we extend its symbols and grammar.

Spatial reasoning in VLMs. Controlled tests show that VLMs answer spatial prepositions far below human accuracy [6], a failure linked to how they bind features to objects [7]. SpatialVLM [16] improves this by training on synthetic spatial data. Open-vocabulary detectors such as Grounding DINO [17] offer boxes from a second model; we instead ask the VLM that already names the objects, so that names and boxes come from one call.

Shortcuts. CLEVR-Hans [9] plants confounders in its training and validation splits (for example, the large cube of CLEVR-Hans3 class 1 is always gray) and removes them in

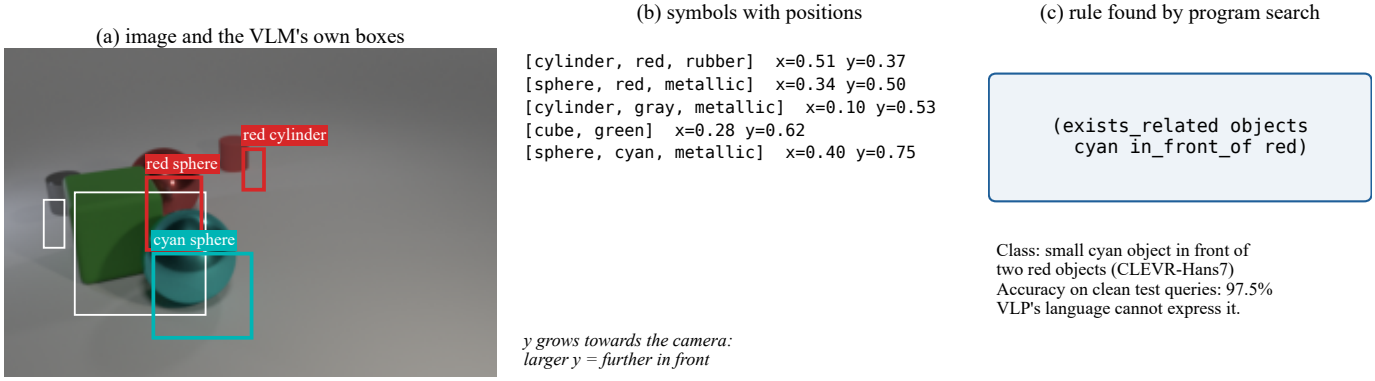


Fig. 1. Object-centric VLP on a CLEVR-Hans7 test image. (a) Qwen2.5-VL-7B's own boxes (image fed at twice its size); they sit slightly low, but their order is right. (b) The symbols the program search sees: VLP's word lists plus box centres. (c) The rule found for this class from 10 positive and 10 negative training images.

the test split, which makes it a direct test of whether a learned rule is the true one.

III. Method

A. Background: Vision-Language Programs

Given positive support images $\mathcal{P}$ and negative ones $\mathcal{N}$, VLP [8] (i) asks the VLM for the task's object and property vocabularies, (ii) calls `get_objects` on every image, which returns a list of entries [*name*, *property*, ...], and (iii) builds a typed domain-specific language (DSL) whose functions include `exists_object`, `exists_property`, `exists_object_with_property` and the Boolean operators. The DSL becomes a probabilistic context-free grammar in which every rule is uniform except the discovered symbols, which are weighted by how often and how specifically they occur in positive images. Heap search [18] enumerates programs in order of probability for a fixed wall-clock budget; the program with the highest training accuracy, ties broken by probability, labels the query images.

B. Object-centric symbols

We keep VLP's `get_objects` prompt and change two things: every object must be listed separately, and each must carry a box in the JSON form Qwen2.5-VL was trained to produce (`"bbox_2d"`) [19]. Boxes come back in the pixel frame of the model's resized input (sides rounded to multiples of 28); we normalise them and take the box centre (x, y), with x to the right and y downwards. In CLEVR's camera view a larger y means closer to the camera, that is, further *in front*. The position is stored as an attribute of the entry and never as a word, so it cannot leak into the vocabulary.

At the native 480×320 size the boxes were stretched vertically (Fig. 2, left). On one development task we found that feeding the image at twice its size, a 952×616 model frame, fixes this (Fig. 2, right; Table III). All object-centric answers use the doubled size; VLP's own calls keep the native size, as released.

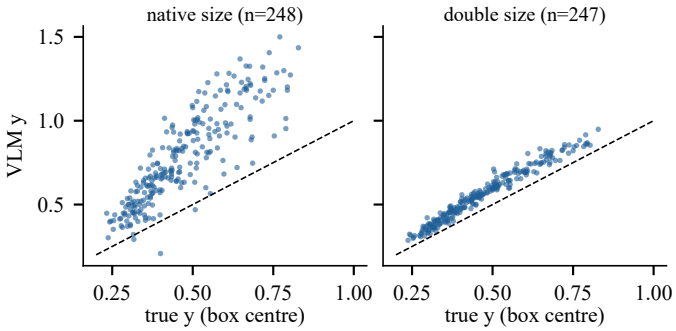


Fig. 2. Vertical box centre returned by the VLM against the true centre, for the same 60 development images fed at native and at double size (attribute-matched objects). Doubling removes the stretch; a small constant offset remains, which leaves the order of objects intact.

C. Spatial grammar

We add three groups of typed functions over the object list O of an image:

- **Counting:** `count_object`(o), `count_property`(p), `count_object_with_property`(o, p), the integers 0 to 6, and the comparisons `gt?` and `eq?`.
- **Regions:** for $R \in \{\text{left, right, front, back}\}$ half of the image, `count_object_in_region`(o, R), `count_property_in_region`(p, R) and `count_object_with_property_in_region`(o, p, R); for example, left$(a) \Leftrightarrow x_a < 0.5$.
- **Relations:** for $\rho \in \{\text{left of, right of, in front of, behind}\}$, with in_front_of$(a, b) \Leftrightarrow y_a > y_b$ and the others defined alike,

$$\text{count_related}(p_1, \rho, p_2) = \max_{a \in O:\, p_1 \in a} \big|\{b \in O \setminus \{a\} : p_2 \in b,\ \rho(a, b)\}\big|, \quad (1)$$

and `exists_related` is true when this count is positive. The maximum over a lets one program state *one cyan object in front of two red ones*.

D. A prior for the added functions

A larger grammar spreads the fixed 10-second search over more programs. In a pilot with ground-truth scenes the full grammar missed the rule VLP's grammar finds on CLEVR-Hans3 class 3 and settled on a spurious one (76.7% against 95.0%). We therefore multiply the weight $w(r)$ of every production r that uses an added function by λ and renormalise over the productions of the same non-terminal A:

$$P(r) = \frac{\lambda^{[r \in E]}\, w(r)}{\sum_{r' \in R(A)} \lambda^{[r' \in E]}\, w(r')}, \tag{2}$$

where E is the set of productions with added functions. With $\lambda < 1$ the search tries VLP's rules first and the new predicates only when they pay. We chose $\lambda = 0.1$ on 20 development tasks with ground-truth scenes (counting / full grammar: $\lambda = 1$: 89.1 / 89.6%, 0.3: 90.6 / 90.7%, 0.1: 90.7 / 91.0%, 0.03: 88.5 / 88.6%) and fixed it for every later run.

IV. Experimental Setup

Data. CLEVR-Hans3 and CLEVR-Hans7 [9] render 3D scenes of cubes, spheres and cylinders. CLEVR-Hans7 classes 3, 5 and 6 are spatial: *a small cyan object in front of two red objects*, *three spheres in the left half*, and *three metal cylinders in the right half*. Every task is scored on queries from the confounded validation split, as in VLP's protocol, and on queries from the unconfounded test split.

Protocols. On CLEVR-Hans3 we reproduce VLP's released loader (10 positive and 10 negative training images as support, 10 + 10 validation queries per class) and add 10 + 10 test queries. On CLEVR-Hans7 we draw three random task sets: for each class, 10 positive and 10 negative support images from training images with id below 1500 (negatives spread over the other classes), plus 20 + 20 validation and 20 + 20 test queries, giving 21 tasks. Every design choice (λ, the image size) was made on development tasks drawn from training ids of 1500 and above.

Conditions. Each row changes one thing: *direct prompting* (VLP's baseline: the VLM states a rule from the support images, then judges each query); *VLP* as released; our prompt with positions hidden and VLP's grammar; + *counting*; and + *geometry*, the full method. The three grammars run on ground-truth scenes as upper bounds.

Model and compute. All conditions use Qwen2.5-VL-7B-Instruct [19] in 16-bit precision on a free Kaggle session with two T4 GPUs. Its default temperature (10^{-6}) makes decoding effectively greedy, so we vary the image draws instead of seeds. The 7,510 VLM answers were cached once and replayed on a laptop CPU, where VLP's own search code runs with its 10-second budget and depth limit 6. Prompts asked in two separate GPU sessions returned identical answers (293 of 293).

Metric and tests. We report balanced accuracy on the query images, averaged over tasks. All images of a task are judged by one program, so tasks, not images, are the independent units: we test paired differences with an exact two-sided sign-flip permutation test over tasks.

TABLE I
CLEVR-Hans7, 21 tasks: mean balanced accuracy (%), ± standard deviation over the three draws. "Test" queries are unconfounded.

	Val queries	Test queries		
Condition	all	all	spatial	other
Direct prompting	51.9 ±7.1	53.0 ±3.4	53.6	52.5
VLP [8]	72.9 ±4.4	69.5 ±2.1	65.6	72.5
Our prompt, VLP grammar	74.5 ±6.5	69.2 ±5.0	62.5	74.2
+ counting	80.1 ±5.0	75.8 ±2.4	77.5	74.6
+ geometry (ours)	**82.6** ±4.0	**80.2** ±5.2	**90.8**	72.3
Ground-truth scenes (upper bounds)				
VLP grammar	81.8	74.9	63.9	83.1
+ counting	86.9	82.7	82.5	82.9
+ geometry	89.8	85.7	89.4	82.9

V. Results

Main result. Table I gives the CLEVR-Hans7 results. On unconfounded test queries our method raises accuracy from 69.5% to 80.2% (+10.7 points; 13 tasks better, 6 worse; $p = 0.013$). On the nine spatial tasks the gain is +25.3 points (8 better, 1 worse; $p = 0.008$), and on the twelve other tasks accuracy is unchanged (72.5% against 72.3%). Validation queries show the same pattern (72.9% to 82.6%, $p = 0.028$). Direct prompting stays near chance, and our system with the VLM's own perception even beats VLP's grammar run on perfect scenes (80.2% against 74.9%).

One change at a time. Our prompt alone changes nothing (-0.4 points on test). Counting adds 6.7 points (5 tasks better, none worse; $p = 0.062$), almost all on classes 5 and 6, where "three spheres" or "three cylinders" approximates the rule without positions. Geometry then adds 13.3 points on the spatial tasks (5 better, none worse; $p = 0.062$, the smallest value five non-zero differences allow) and costs 2.3 points on the others: two tasks got worse, one by fitting a spurious relation (*green in front of purple*), the other because the larger grammar changed which equally fitting rule the search reached first. Fig. 3 shows the classes: class 3 rises from 66.7% to 89.2%, class 5 from 82.5% to 91.7%, class 6 from 83.3% to 91.7%.

Why the VLM version beats true scenes on class 3. With exact symbols, the planted confounder of class 3 (its cyan object is always a small cube in training) fits the support set as well as the true rule, and the shorter program `exists_object_with_property(cube, cyan)` wins in two of three draws; it drops to 65% and 73% on clean test images. With the VLM's symbols the relation rule won in two of three draws (100% and 97.5% on the test split). Program search finds the simplest rule consistent with 20 examples; when a shortcut is equally consistent, nothing in the search prefers the true rule.

CLEVR-Hans3. Our run of VLP reproduces the published number (86.7% against 86.1%; Table II). Geometry costs one task on validation queries and nothing on the test split, where all object-centric conditions reach 76.7% against VLP's 60.0%. The gap shows how much the confounded protocol hides: VLP's

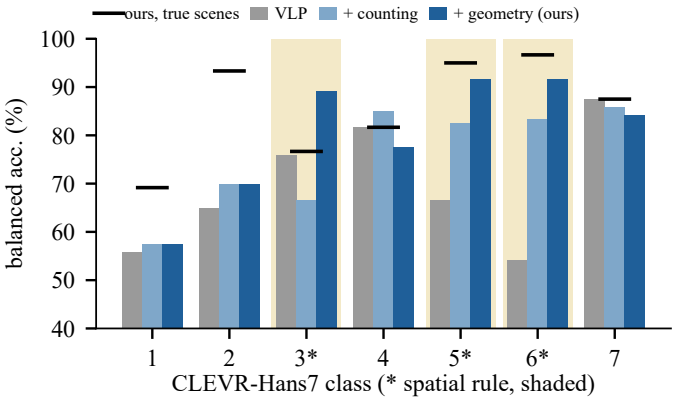


Fig. 3. Test-split accuracy per CLEVR-Hans7 class (mean of three draws). The gain from geometry is confined to the spatial classes.

TABLE II
CLEVR-HANS3 WITH VLP'S OWN PROTOCOL (3 TASKS; NO SPATIAL RULES).

Condition	Val queries	Test queries
Direct prompting	50.0	56.7
VLP (paper reports 86.1 on val)	86.7	60.0
Our prompt, VLP grammar	86.7	76.7
+ counting	86.7	76.7
+ geometry (ours)	83.3	76.7

rule for class 1 is `(and (exists_object_with_property cube gray) (exists_object cylinder))`, which uses the planted gray color and falls from 90% on validation queries to 60% on test queries.

Perception. Table III measures the boxes on objects whose shape and color are unique in an image. At double size the order of two objects is right 98% to 99% of the time; absolute halves are weaker (77%) because of the constant offset in Fig. 2. Our prompt also counts objects better than VLP's own prompt (mean absolute error 0.28 against 0.72 objects per image).

VI. Discussion and Limitations

Positions help where rules are spatial and are close to neutral elsewhere; the prior λ keeps the larger grammar from crowding out VLP's rules, although two of twelve non-spatial tasks still got worse, one through a spurious relation fitted to 20 examples. Several limits remain. The study is small: 21 CLEVR-Hans7 tasks of which nine are spatial, so the geometry step alone reaches only $p = 0.062$ at task level. With this VLM, VLP's vocabulary discovery never proposed size words, so the size part of classes 1, 2, 4 and 7 is out of reach for every VLM condition; the ground-truth rows show the headroom. Regions depend on absolute positions and inherit the box offset, while relations depend only on order and are robust. All results use one VLM and synthetic scenes, and the wall-clock search budget makes exact numbers hardware-dependent. Natural-image benchmarks with spatial concepts, depth cues beyond image height, and a search that prefers rules surviving held-out examples are the next steps.

TABLE III
QUALITY OF THE VLM'S OBJECT LISTS AGAINST GROUND TRUTH. LEFT: THE SAME 60 DEVELOPMENT IMAGES AT NATIVE AND DOUBLE SIZE. RIGHT: ALL 2,700 OBJECT-CENTRIC ANSWERS (DOUBLE SIZE).

Measure	native	double	all, double
Position correlation, x / y	0.93 / 0.87	0.99 / 0.98	0.99 / 0.98
Vertical scale (fitted slope)	1.72	1.10	1.14
Left/right order of two objects	98.0%	99.6%	99.3%
Front/behind order of two objects	96.9%	98.9%	98.0%
Front-half agreement	56.5%	79.8%	77.1%
Mean absolute count error			0.28

VII. Conclusion

Giving each object a position, taken from the VLM's own boxes, and adding counting, region and relation predicates lets Vision-Language Programs learn spatial rules they could not express before. On CLEVR-Hans7 this lifts unconfounded test accuracy from 69.5% to 80.2%, and from 65.6% to 90.8% on spatial classes, with no training and no new model. The method keeps VLP's main strength, a rule a person can read and check, which is also how we found that both systems sometimes learn the dataset's shortcuts.

Code and Data

Our code builds on the released VLP code [20] and the CLEVR-Hans data [21] (MIT licence). Our code is available at https://github.com/bukhari-hamzamukhtar/OCVLP.

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